ASSUMPTION UNIVERSITY OF THAILAND

Network Design and Security Configuration TechSafe Ltd Infrastructure Implementation

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***Abstract*—This project presents the complete network design and security configuration for TechSafe Ltd. The objective was to design a structured network using subnetting, analyze HTTP/HTTPS traffic using Wireshark, and implement firewall security rules using Microsoft Windows Defender Firewall. Cisco Packet Tracer was used for network topology design, Wireshark for traffic analysis, and Windows Defender Firewall for security enforcement. The project demonstrates practical implementation of IP subnetting, traffic filtering, and access control mechanisms.**

***Index Terms*—Network Design, Subnetting, Cisco Packet Tracer, Wireshark, Firewall, HTTP, HTTPS, FTP, Security Configuration**

I. Introduction

Modern organizations require structured network architectures combined with robust security policies. This project focuses on designing and securing TechSafe Ltd's internal network. The implementation includes subnet segmentation, traffic monitoring, and firewall rule enforcement to restrict unauthorized services.

II. Technologies Used

- **Cisco Packet Tracer** – Network topology design and subnet implementation
- **Wireshark** – Network traffic monitoring and HTTPS analysis
- **Microsoft Windows Defender Firewall** – Firewall rule configuration

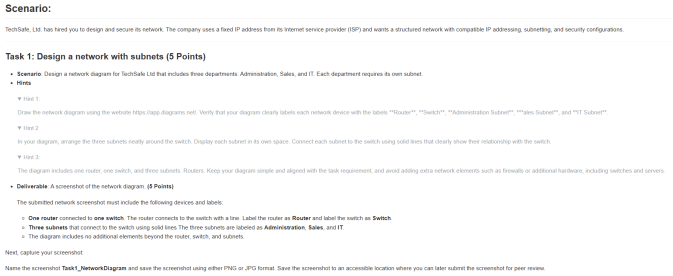
Fig. 1. Technologies Used in the Project

III. Task 1: Network Design with Subnets

The network was divided into three departments:

- Administration
- Sales
- IT

Each department was assigned its own subnet. A single router connects to one switch, and each department connects to the switch.



Scenario:

TechSafe_Ltd. has hired you to design and secure its network. The company uses a fixed IP address from its Internet service provider (ISP) and wants a structured network with compatible IP addressing, subnetting, and security configurations.

Task 1: Design a network with subnets (5 Points)

- **Scenario**: Design a network diagram for TechSafe Ltd that includes three departments: Administration, Sales, and IT. Each department requires its own subnet.
- **Hints**

  ▼ Hint 1

  Draw the network diagram using the website https://app.diagrams.net/. Verify that your diagram clearly labels each network device with the labels "Router", "Switch", "Administration Subnet", "Sales Subnet", and "IT Subnet".

  ▼ Hint 2

  In your diagram, arrange the three subnets neatly around the switch. Display each subnet in its own space. Connect each subnet to the switch using solid lines that clearly show their relationship with the switch.

  ▼ Hint 3

  The diagram includes one router, one switch, and three subnets. Routers. Keep your diagram simple and aligned with the task requirement, and avoid adding extra network elements such as firewalls or additional hardware, including switches and servers.

- **Deliverable**: A screenshot of the network diagram. **(5 Points)**

  The submitted network screenshot must include the following devices and labels:

  - **One router** connected to **one switch**. The router connects to the switch with a line. Label the router as **Router** and label the switch as **Switch**.
  - **Three subnets** that connect to the switch using solid lines The three subnets are labeled as **Administration**, **Sales**, and **IT**.
  - The diagram includes no additional elements beyond the router, switch, and subnets.

Next, capture your screenshot:

Name the screenshot **Task1_NetworkDiagram** and save the screenshot using either PNG or JPG format. Save the screenshot to an accessible location where you can later submit the screenshot for peer review.

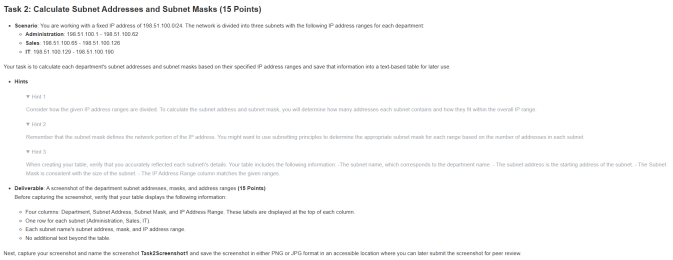
Fig. 2. Network Topology Designed Using Cisco Packet Tracer

IV. Task 2: Subnet Address Calculation

The assigned network was: 198.51.100.0/24

The network was divided into three /26 subnets:

- Administration: 198.51.100.0/26
- Sales: 198.51.100.64/26
- IT: 198.51.100.128/26



Task 2: Calculate Subnet Addresses and Subnet Masks (15 Points)

- **Scenario**: You are working with a fixed IP address of 198.51.100.0/24. The network is divided into three subnets with the following IP address ranges for each department:
  - **Administration**: 198.51.100.1 - 198.51.100.62
  - **Sales**: 198.51.100.65 - 198.51.100.126
  - **IT**: 198.51.100.129 - 198.51.100.190

Your task is to calculate each department's subnet addresses and subnet masks based on their specified IP address ranges and save that information into a text-based table for later use.

- **Hints**

  ▼ Hint 1

  Consider how the given IP address ranges are divided. To calculate the subnet address and subnet mask, you will determine how many addresses each subnet contains and how they fit within the overall IP range.

  ▼ Hint 2

  Remember that the subnet mask defines the network portion of the IP address. You might want to use subnetting principles to determine the appropriate subnet mask for each range based on the number of addresses in each subnet.

  ▼ Hint 3

  When creating your table, verify that you accurately reflected each subnet's details. Your table includes the following information: - The subnet name, which corresponds to the department name. - The subnet address is the starting address of the subnet. - The Subnet Mask is consistent with the size of the subnet. - The IP Address Range column matches the given ranges.

- **Deliverable**: A screenshot of the department subnet addresses, masks, and address ranges **(15 Points)**

  Before capturing the screenshot, verify that your table displays the following information:

  - Four columns: Department, Subnet Address, Subnet Mask, and IP Address Range. These labels are displayed at the top of each column.
  - One row for each subnet (Administration, Sales, IT).
  - Each subnet name's subnet address, mask, and IP address range.
  - No additional text beyond the table.

Next, capture your screenshot and name the screenshot **Task2Screenshot1** and save the screenshot in either PNG or JPG format in an accessible location where you can later submit the screenshot for peer review.

Fig. 3. Subnet Address Table

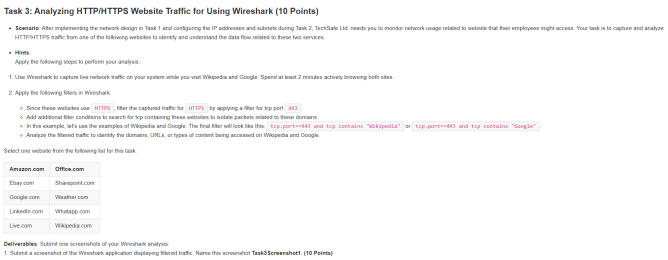
Each subnet contains 64 IP addresses (62 usable hosts).

V. Task 3: HTTP/HTTPS Traffic Analysis Using Wireshark

Wireshark was used to capture HTTPS traffic while browsing Google. A filter was applied:

```
tcp.port == 443 and tcp contains "google"
```

This allowed isolation of encrypted HTTPS packets.



Task 3: Analyzing HTTP/HTTPS Website Traffic for Using Wireshark (10 Points)

- **Scenario**: After implementing the network design in Task 1 and configuring the IP addresses and subnets during Task 2, TechSafe Ltd. needs you to monitor network usage related to website that their employees might access. Your task is to capture and analyze HTTP/HTTPS traffic from one of the following websites to identify and understand the data flow related to these two services.
- **Hints**
  Apply the following steps to perform your analysis:

1. Use Wireshark to capture live network traffic on your system while you visit Wikipedia and Google. Spend at least 2 minutes actively browsing both sites.
2. Apply the following filters in Wireshark:
   - Since these websites use `HTTPS`, filter the captured traffic for `HTTPS` by applying a filter for tcp port `443`.
   - Add additional filter conditions to search for tcp containing these websites to isolate packets related to these domains.
   - In this example, let's use the examples of Wikipedia and Google. The final filter will look like this: `tcp.port==443 and tcp contains "Wikipedia"` or `tcp.port==443 and tcp contains "Google"`
   - Analyze the filtered traffic to identify the domains, URLs, or types of content being accessed on Wikipedia and Google.

Select one website from the following list for this task.

Amazon.com	Office.com
Ebay.com	Sharepoint.com
Google.com	Weather.com
LinkedIn.com	Whatapp.com
Live.com	Wikipedia.com

Deliverables: Submit one screenshots of your Wireshark analysis:
1. Submit a screenshot of the Wireshark application displaying filtered traffic. Name this screenshot **Task3Screenshot1**. **(10 Points)**

Fig. 4. Wireshark HTTPS Traffic Analysis

The analysis confirmed TLS handshake and encrypted communication over port 443.

VI. Task 4: Firewall Configuration Using Windows Defender Firewall

Security policies were implemented to:

- Block FTP traffic (Port 21)
- Block HTTP traffic (Port 80)

A. FTP Access Before Blocking

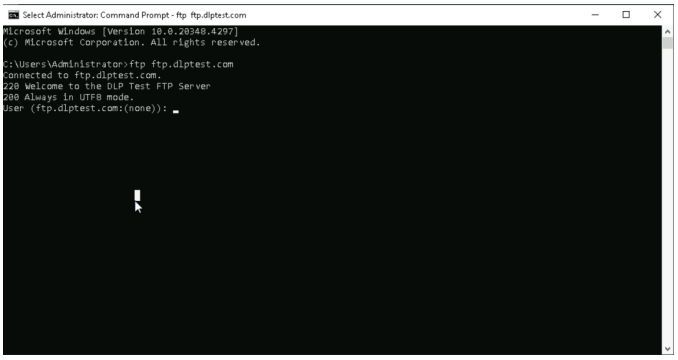


Fig. 5. Successful FTP Access Before Firewall Rule

B. FTP Rule Configuration

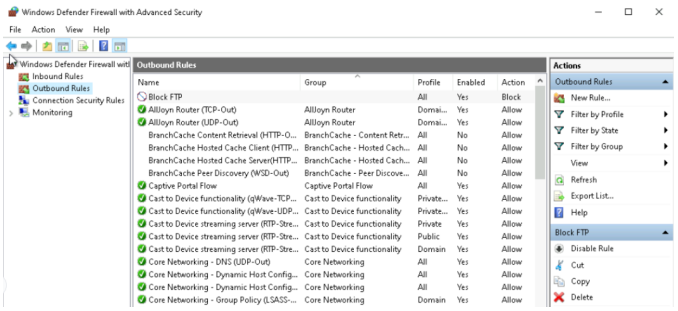
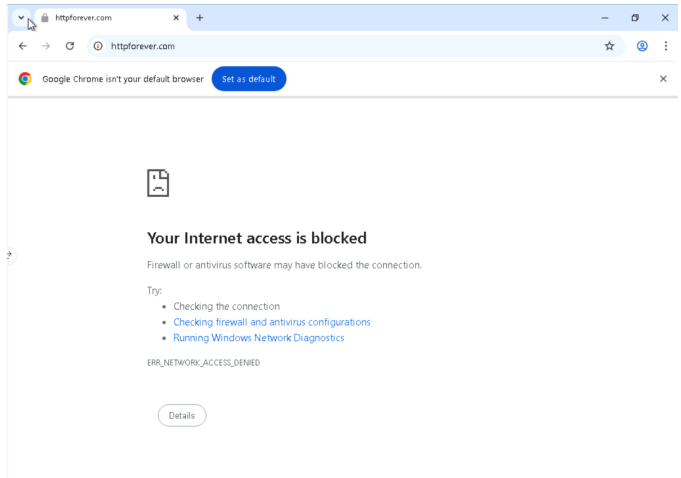


Fig. 6. Outbound Rule Blocking FTP

C. FTP Access After Blocking

Fig. 7. FTP Access Blocked

D. HTTP Blocking Verification



VII. Security Analysis

The implementation demonstrates:

- Logical network segmentation improves isolation
- HTTPS encrypts data (TLS protocol)
- Firewall rules effectively restrict unauthorized services
- Port-based filtering enhances network security posture

VIII. Conclusion

The project successfully designed and secured TechSafe Ltd's network infrastructure. Subnetting improved network organization, Wireshark enabled traffic visibility, and firewall rules enforced strict access control. This demonstrates practical knowledge of network architecture and cybersecurity implementation in enterprise environments.